



AUGUST 16 2024

Chemical Engineering Faculty Retreat Undergraduate Program

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UNIVERSITY OF
ILLINOIS CHICAGO



UIC Chemical Engineering Undergraduate Program

- Overview
- Program Improvements
 - Program revisions
 - New elective courses
 - ChE and Data Science
- ABET is coming → Fall 2026
 - Student Outcome Assessment in progress
- Alumni relations
- External Advisory Board



ChE Curriculum

Note: CHE 150 Climate Engineering is a General Education course offered to Non-ChE majors

Core Courses

Thermodynamics-Based Courses

ChE 201 – Introduction to thermodynamics
ChE 301 – Chemical Engineering Thermodynamics
ChE 321 – Chemical Reaction Engineering

Process-Oriented Courses

ChE 210 – Material & Energy Balances
ChE 341 – Chemical Process Control
ChE 396 – Senior Design I (Equipment Design)
ChE 397 – Senior Design II (Design teams)

Numerical Analysis

ChE 205- Numerical Methods in ChE

Engineering Education & Development

ENGR100 -Engineering Orientation (Engineering Success Program, Guaranteed Paid Internship Program)
ChE 499- Professional Development Seminar

Transport-Based Courses

ChE 311 – Transport Phenomena I (Fluid Mechanics)
ChE 312 – Transport Phenomena II (Heat & Mass Transfer)
ChE 313 – Transport Phenomena III (Separations)

Unit Operations Laboratory

ChE 381 – Chem Eng Lab I
ChE 382 – Chem Eng Lab II

Molecular Engineering Courses

CHE 230 Molecular Engineering in CHE
CHE 330 Polymer Science

ChE 400-level Elective
CHE 392:
Undergraduate
Research

Six Concentrations:

Biochemical Engineering

Energy & Environment

Process Automation

Nanotechnology

Polymers & Molecular Engineering

Entrepreneurship





ChE 400-level courses: Undergraduate and Graduate students

CHE 410 Transport Phenomena

CHE 413 Introduction to Flow in Porous Media

CHE 421 Combustion Engineering

CHE 422 Biochemical Engineering

CHE 423 Catalytic Reaction Engineering

CHE 425 Nanotechnology for Pharmaceutical Applications

CHE 427 Entrepreneurship in Engineering

CHE 433 Process Simulation w/Aspen+

CHE 438 Computational Molecular Modeling

CHE 440 Non-Newtonian Fluids

CHE 441 Computer Applications in Chemical Engineering

CHE 451 Renewable Energy Technology

CHE 450 Air Pollution

CHE 453 Electrochemistry

CHE 454 Molecular & Macromolecular Engineering

CHE 455 Nanoscale Systems in Chemical Engineering

CHE 456. Fundamentals and Design of Microelectronics Processes

CHE 457 Colloidal and Interfacial Phenomena

CHE 494 Clean Energy & the Electric Grid

CHE 494 Biopharmaceutical Process Engineering



Make permanent courses?



Required ChE electives for the concentrations

Energy & Environment

CHE 451 Renewable Energy Technology and

CHE 450 Air Pollution or

CHE 453 Electrochemistry or

400-level course approved by DUGS or
CHE 392 Undergraduate Research



CHE 494 Clean Energy & the Electric
Grid used as CHE elective

Biochemical Engineering

CHE 422 Biochemical Engineering



CHE 494 Biopharmaceutical Process
Engineering used as CHE elective

Process Automation

CHE 433 Process Simulation w/Aspen+



Concentrations and their required ChE electives

Polymers & Molecular Engineering

CHE 454 Molecular & Macromolecular Engineering and choose between:

CHE 425 Nanotechnology for Pharmaceutical Applications

CHE 438 Computational Molecular Modeling

CHE 440 Non-Newtonian Fluids

CHE 453 Electrochemistry

CHE 455 Nanoscale Systems in Chemical Engineering

CHE 457 Colloidal and Interfacial Phenomena

400-level course approved by the Director of Undergraduate Studies

CHE 392 Undergraduate research

Nanotechnology

CHE 455 Nanoscale Systems in Chemical Engineering and choose between:

CHE 425 Nanotechnology for Pharmaceutical Applications

CHE 457 Colloidal and Interfacial Phenomena

400-level course approved by the Director of Undergraduate Studies

CHE 392 Undergraduate research

Entrepreneurship

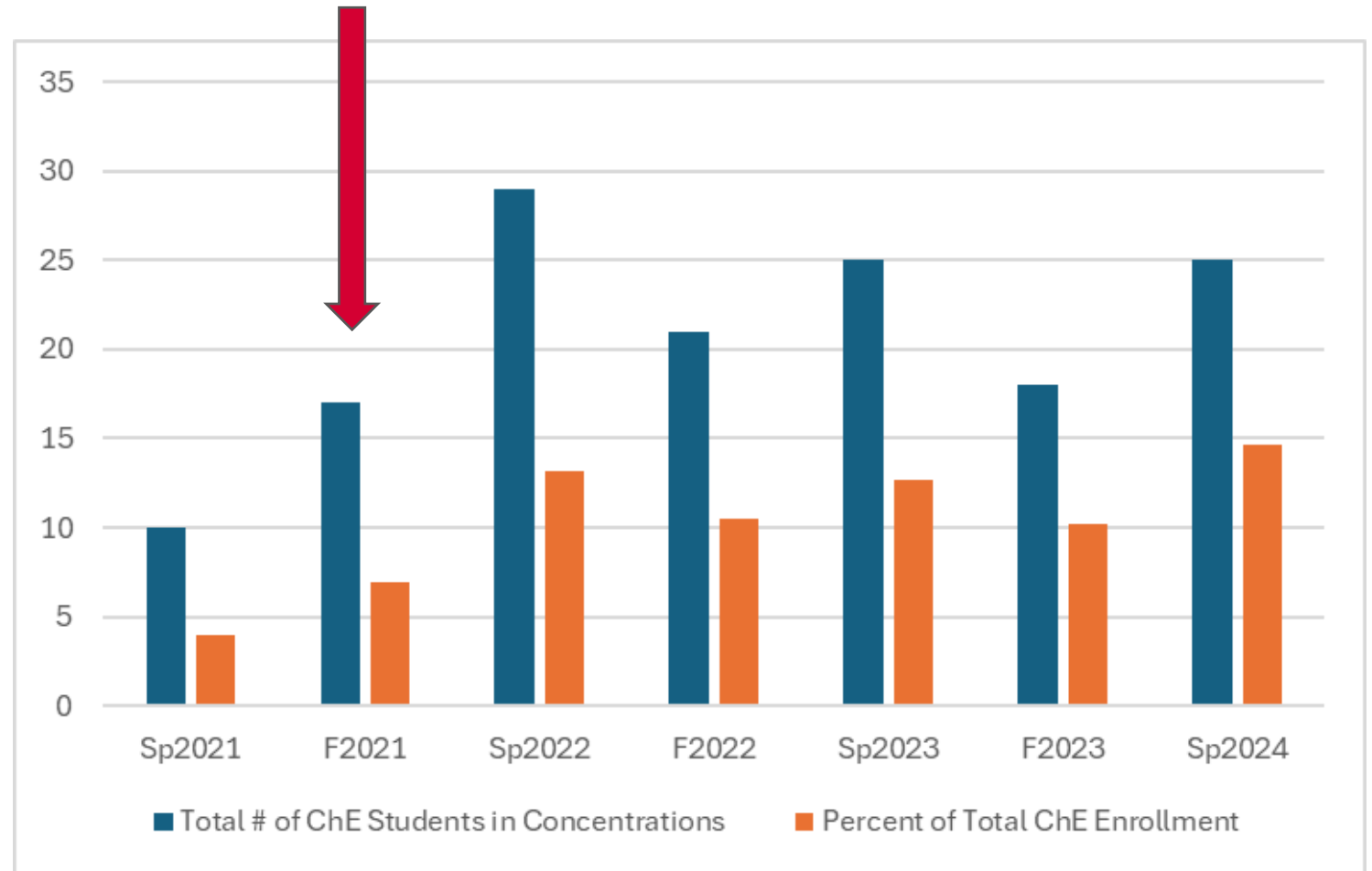
CHE 427 Entrepreneurship in Engineering



Students taking a concentration:

Spring 2021 to Spring 2024

New concentrations added
Fall 2021

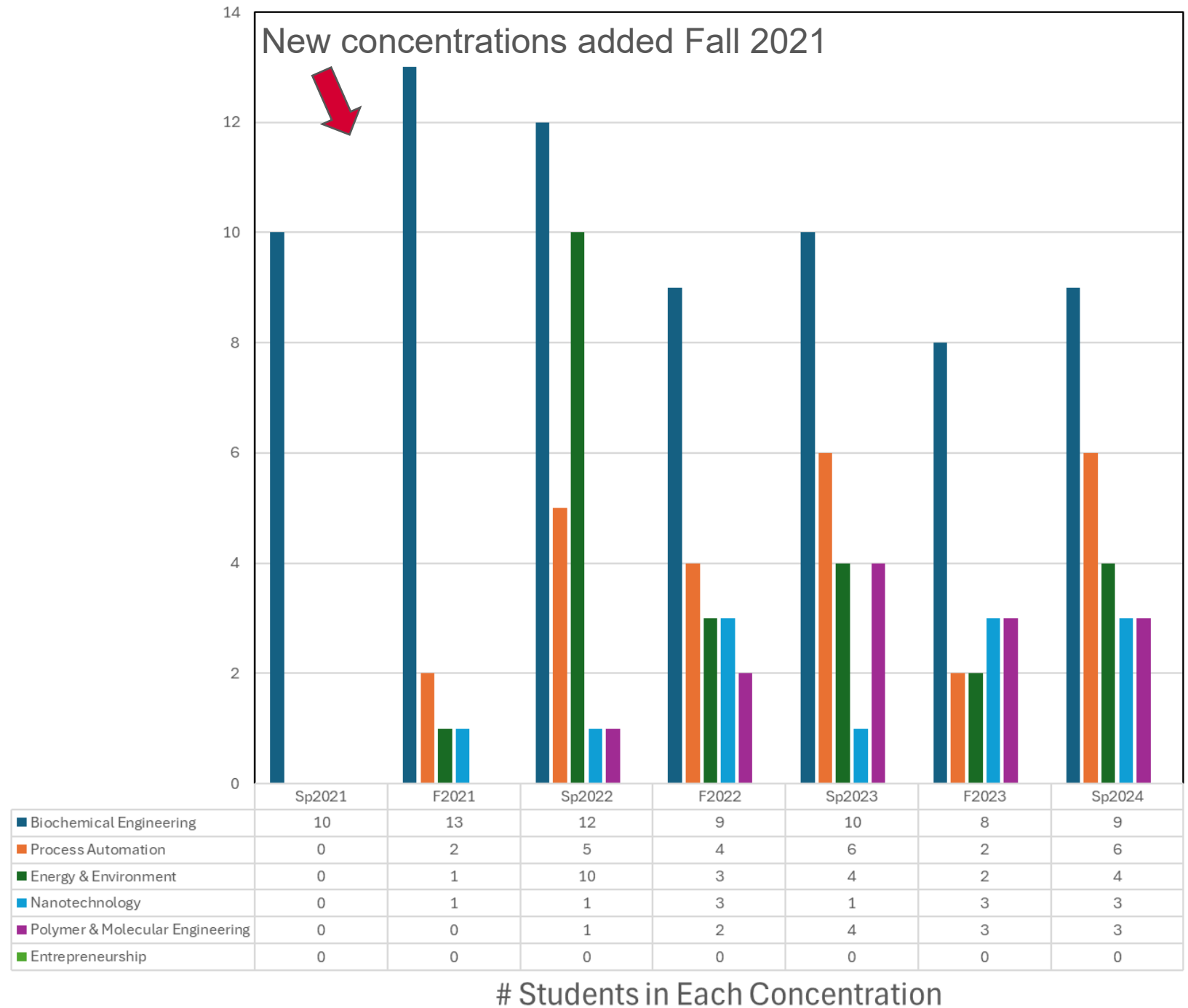


Biochemical Engineering, Energy & Environment,
Entrepreneurship, Nanotechnology, Polymers and
Molecular Engineering, Process Automation





Number of students in each concentration: Spring 2021 to Spring 2024





Minors Taken by ChE's (20) AY 2021 to 2024

ART

BIOMEDICAL ENGINEERING

BIOLOGICAL SCIENCES

BUSINESS ADMINISTRATION

BUSINESS ANALYTICS

CHEMISTRY(?)

CHINESE

COMPUTER SCIENCE (2-6 STUDENTS)

CRIMINOLOGY, LAW, AND JUSTICE

DISABILITY & HUMAN DEVELOPMENT

FINANCE (4 STUDENTS)

FRENCH AND FRANCOPHONE STUDIES

INDUSTRIAL ENGINEERING

MANAGEMENT INFORMATION SYSTEMS

MATERIALS ENGINEERING

MATH & COMPUTER SCIENCE (2-4 STUDENTS)

MATHEMATICS (6-10 STUDENTS)

PHILOSOPHY

PSYCHOLOGY

RUSSIAN STUDIES



Curriculum Improvement Discussion: Program Revisions

PROGRAM REVISION PROPOSALS:

(1)

ADD CHE 101 INTRODUCTION TO
SUSTAINABLE CHEMICAL ENGINEERING
TO CORE CURRICULUM (+3 HOURS)

CREATE A NEW SELECTIVE OPTION: CHE
230 MOLECULAR SYSTEMS IN CHEMICAL
ENGINEERING OR CME 260 PROPERTIES
OF MATERIALS (-3 HOURS)

(2)

REMOVE CHEM 234 CHEMICAL SYNTHESIS
(ORG II) FROM MATH/SCIENCE
REQUIREMENT (-3 HOURS)

ADD SECOND TECHNICAL ELECTIVE TO
CORE REQUIREMENT (+3 HOURS)



Curriculum Improvement Discussion:

Technical electives

NEW TECHNICAL ELECTIVE COURSES TO CONSIDER:

PROCESS EQUIPMENT DESIGN AND SAFETY

CHEMICAL TECHNOLOGY



Curriculum Improvement Discussion: Chemical Engineering & Data Science

CHEMICAL ENGINEERING AND DATA SCIENCE

HOW CAN WE ADD DATA SCIENCE TO OUR
CURRICULUM?

(1)

BS IN CHE WITH DATA SCIENCE
CONCENTRATION

(UIUC IS IMPLEMENTING A SIMILAR
SPECIALIZATION)

(2)

BS IN DATA SCIENCE WITH CHE
CONCENTRATION

(UIC COMPUTER SCIENCE DEPARTMENT HAS
SEVERAL PROGRAMS WITH OTHER MAJORS)





ABET Assessment

7 Student Learning Outcomes

Students graduating from the chemical engineering program at UIC will have:

SO	Description
1	An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.
2	An ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors
3	An ability to communicate effectively with a range of audiences.
4	An ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.
5	An ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.
6	An ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.
7	An ability to acquire and apply new knowledge as needed, using appropriate learning strategies.

Student Learning Outcome Course Rubric Example

ChE 205- Computational Methods in Chemical Engineering- Spring 2022

Student Learning Outcome 1

Rubric for assessment for Final exam, Problems, 1c, 2, 4, and 3a

RESULTS BY PERFORMANCE INDICATOR				
Rubric Item (PI)	1	2	4	6
Mapped SLO	1	1	1	1
Average	2.25	2.25	2.10	2.85
Std Dev	0.79	0.64	0.79	0.37

RESULTS BY STUDENT OUTCOME	
SO	1
Average Score	2.36
Standard dev.	0.29

Analysis:

Average scores were fairly uniform across Performance Indicators (PI) 1, 2, and 4. The strongest score was for Performance Indicator 6; Using appropriate resources to solve problems. A lower score was observed for PI 4, Translating academic theory into engineering applications. There was a fairly even distribution between satisfactory and developing scores for PI's 1, 2 and 4. This data suggests that most students were strong in (i) using appropriate resources needed to solve problems, were developing in (ii) applying mathematical and scientific principles to formulate models, and (iii) solving engineering problems, but were weak in (iv) translating academic theory into engineering applications.

Points of Improvement:

- There is a broad range of student skill level in programming and Excel. Most students are still developing these skills and need extra help. It would be beneficial to cover the basics of Excel and programming at the beginning of the course and provide resources on those topics.

SLO 1 Performance Indicator (PI)	(1) Unsatisfactory	(2) Developing	(3) Satisfactory
1. Apply mathematical and scientific principles to formulate models and systems relevant to the subject	Able to successfully combines mathematical and/or scientific principles to formulate models and systems relevant to chemical engineering	Chooses a mathematical model or scientific principle that applies to an engineering problem, but has trouble in model development	Does not understand the connection between mathematical models and the system or process to be analyzed or designed



Student Outcome Assessment Summary Basis: Course Rubrics Senior Exam Senior EXPO

Summary of “S” and “D” attainment for all Student Outcomes:

3 = S = Satisfactory 2 = D = Developing 1 = U = Unsatisfactory

Percentage of students attaining a rating of Satisfactory for all the SO's

SO	2020-2021	2021-2022
1	78.4	73.0
2	72.0	86.0
3	80.3	81.2
4	82.6	84.8
5	72.7	82.9
6	75.1	71.5
7	71.1	85.4

Percentage of students attaining a rating of Satisfactory or Developing for all SO's

SO	2020-2021	2021-2022
1	94.6	90.4
2	97.1	95.3
3	95.9	97.7
4	98.8	96.7
5	97.9	94.8
6	93.9	90.1
7	97.1	98.6



Alumni Relations

ALUMNI EVENT BEING PLANNED THIS SEMESTER

CHE ALUMNI PROVIDE STRONG SUPPORT TO OUR DEPARTMENT ACTIVITIES:

- CHE COURSES: 499 SEMINAR SPEAKERS, 396/397 SENIOR DESIGN MENTORS, EXPO JUDGES
- AICHE STUDENT CHAPTER ACTIVITIES
- FEEDBACK ON CURRICULUM IMPROVEMENTS

~750 ALUMNI IN CONTACT LIST: 2012 TO 2024 UPDATED EVERY SPRING:

UIC CHEMICAL ENGINEERING LINKEDIN GROUP



External Advisory Board (EAB)

EAB MEETING BEING PLANNED THIS SEMESTER

ACTIVE MEMBERS:

DR. TARUN CHANDRA, EMPIRIQA LLC

DR. S. S. KUMARAN, CABOT CORPORATION

MR. DENNIS O'BRIEN, JACOBS ENGINEERING (RET)

DR. ANIL OROSKAR, OROCHEM TECHNOLOGIES

DR. GAVIN TOWLER, HONEYWELL CORPORATE

RECENTLY STEPPED DOWN:

MR. MICHAEL H. GOLUSZKA, LA-CO INDUSTRIES

DR. DIANE GRAZIANO, ARGONNE NATIONAL LAB

DR. MICHAEL SEACRIST, MEMC

NEW ADVISORY BOARD MEMBER NEEDED



Chemical Engineering Recruitment

**EMPHASIZING CHEMICAL ENGINEERS' ROLE IN
SUSTAINABILITY AT OUR RECRUITMENT EVENTS:**

SEPTEMBER 2024 OPEN HOUSE BLURB:

**CHEMICAL ENGINEERING: SHAPING A SUSTAINABLE
FUTURE**

DISCOVER HOW CHEMICAL ENGINEERS ARE
TRANSFORMING OUR FUTURE WITH SUSTAINABLE
SOLUTIONS, FROM RENEWABLE ENERGY, CLEAN WATER
AND ENVIRONMENTAL TECHNOLOGIES TO LIFE-SAVING
DRUGS, SUSTAINABLE FOOD AND ECO-FRIENDLY
PRODUCTS. SEE HOW YOU CAN JOIN THIS EXCITING
FIELD AND MAKE A REAL DIFFERENCE IN THE WORLD!